

Database Project: Phase I

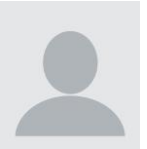
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System Concept

Our initial concept for our project system is a hypothetical basic social media platform. Users can log onto the site using their Username and Password, and can create, like and reply to posts, and follow other users.

Example



From @user

This is a sample post
for my social media
demo!

❤️ 125

#sampletag

#other-sample-tag

This is a reply to the post

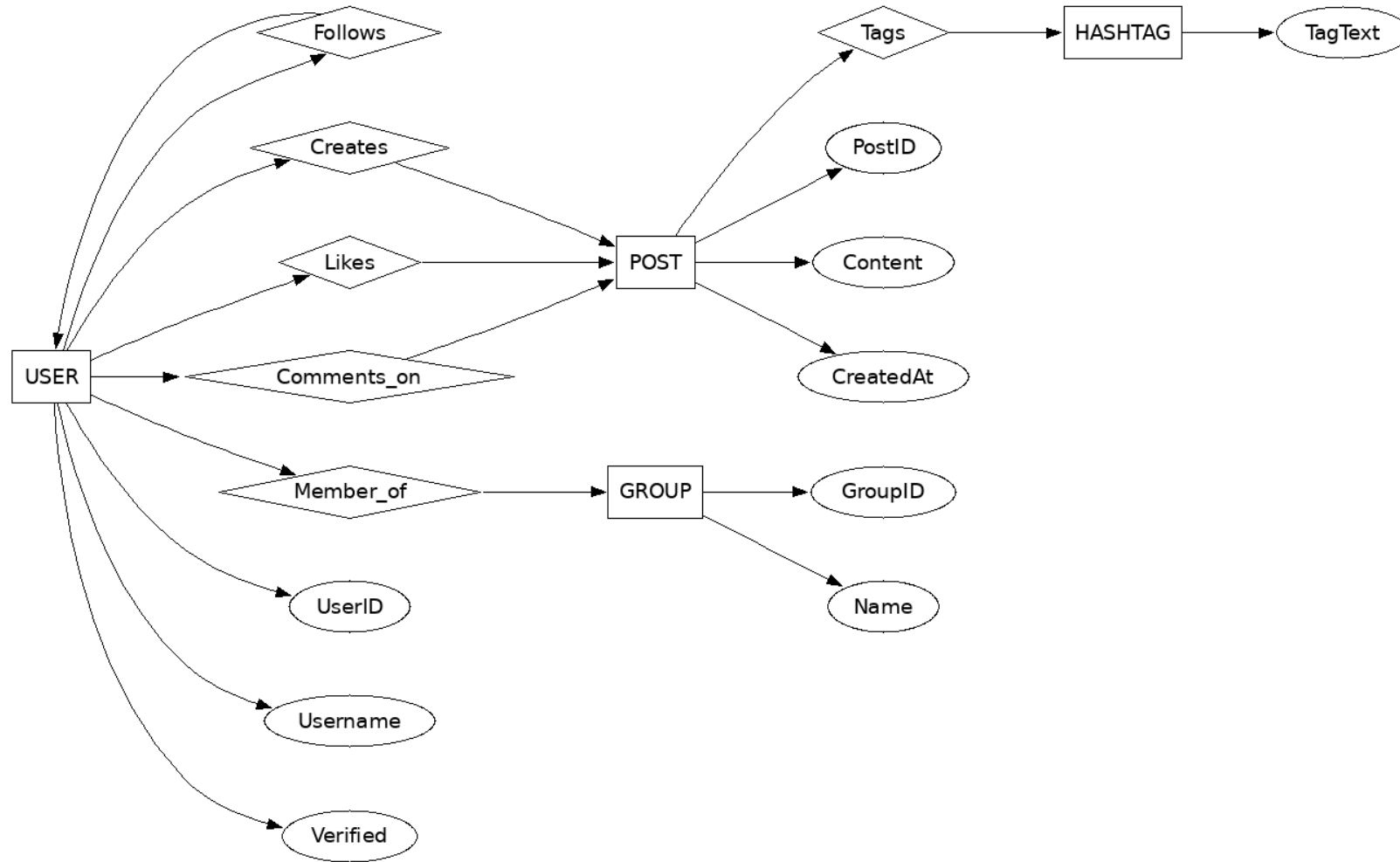
from @other-user

This is a reply to the post... 2!

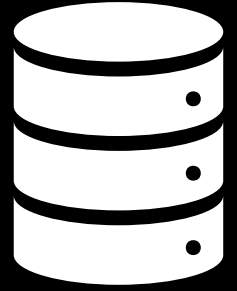
from @other-user

Posts can have tags which group them up with posts of a similar topic, and can additionally mention other users, which will notify them of the post. Posts also have a number ID used in the link, and keep track of the date they were created upon.

ER Diagram



Queries



Some potential queries we will be implementing for our database:

- "Following" tab: return posts of users that a given user is following all in chronological order for a feed-style display.
- "Trending" tab: return most liked posts within a certain time threshold
- "Friends": return list of mutual followers for a given user (list of users that the user is following that are following them back.)

Technology Choice

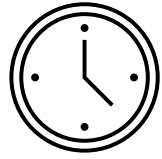
For our frontend: HTML, CSS, JavaScript,

For our Vizualization: D3.js

For our backend: Python(flask) with MySQL-Connector-python,

For our database: MySQL

Project Timeline



- **3/3/26: Phase I Presentation. Complete initial conceptualization for project and the nodes needed for the database design. Set up GitHub repository**
- **3/8/26: Finalize Phase I report**
- **3/26/26: Phase II Presentation. Initial database backend draft and design completed**
- **4/21/26: Phase III Presentation. Basic prototype for platform with basic, limited post and user account viewer frontend**
- **5/7/46: Phase IV Presentation. Final demo of website and completed architecture w/ final report**

The End

Thanks for your time!